

Coordinates, Line Plots and Patterns — Find the Mistake

Answer Key

Grades 4–5

Quick Answers

1. 2	3. 4	5. 2.5	7. 7, 21
2. 10	4. 4	6. 15	8. —

Curriculum

Level: Grades 4–5

4.MD / 5.MD – problems 4, 5, 6

5.G.A.1–A.2 / 5.OA.B.3 – problems 1, 2, 3, 7, 8

Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. If they got 0: decided the two points must be the same place because they are made from the same two numbers
 2. If they got 5: counted only the sideways move and forgot the up-and-down move
 3. If they got 12: answered with the first number of the pair, which is the across value, when the up value was asked for
 4. If they got 3: repeated the height printed under the marks instead of counting the marks stacked above it
 5. If they got 5: answered with how many marks stand in the tallest stack instead of the height that stack sits above
 6. If they got 5: counted how many different heights are marked on the line instead of how many plants were measured
 7. If they got 21: put the term from the second pattern in the first slot of the pair
 7. If they got 7: put the term from the first pattern in the second slot of the pair
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1. Warm-up: from C(5, 4) to D(4, 5) along grid lines.

Step 1: read each pair the same way. C is 5 across and 4 up. D is 4 across and 5 up. Same two numbers, opposite jobs.

Step 2: going from C to D, the across-count drops from 5 to 4, which is 1 step left. The up-count rises from 4 to 5, which is 1 step up.

Step 3: $1 + 1$, so the trip takes 2 steps.

Watch for: the answer 0, from deciding the two points must be the same place because they use the same two numbers. The grid settles it — C and D are plainly two different dots.

2. Find and fix: Nia's (2, 7).

Step 1: name what Nia did. She read the pair as (up, across) instead of (across, up). Counting 7 across and 2 up lands on the point labelled **B**. The point she meant, 2 across and 7 up, is labelled **A**.

Step 2: count the across move from B(7, 2) to A(2, 7): from 7 to 2 is 5 steps left.

Step 3: count the up move: from 2 to 7 is 5 steps up.

Step 4: $5 + 5$, so the two points are 10 steps apart along the grid lines. Swapping the numbers did not move the point a little — it moved it right across the grid.

Watch for: the answer 5, which counts the sideways move only and stops before the up-and-down move.

3. Find and fix: Ravi's pattern pair.

Step 1: build a few pairs so the pattern is visible. Pattern X gives 0, 3, 6, 9, 12 and Pattern Y gives 0, 1, 2, 3, 4, so the pairs are (0, 0), (3, 1), (6, 2), (9, 3), (12, 4).

Step 2: name the rule connecting them. Each X-value is 3 times its Y-value, so the up-value is the across-value divided by 3.

Step 3: the pair with across-value 12 has up-value $12 \div 3$, which is 4. The pair is (12, 4).

Full-credit answer says: “Ravi gave the first number of the pair when the second was asked for.” The across-value 12 is the one he repeated.

4. Find and fix: Omar's 3 plants.

Step 1: name what Omar did. He read the number printed under the marks and used it as his answer. On a line plot that number is the *height*; it never says how many.

Step 2: find the 3 on the line and count the marks stacked above it: there are four.

Step 3: 4 plants are exactly 3 inches tall.

Watch for: the answer 3, which is Omar's — the height copied instead of the marks counted.

5. Which height happened most often?

Step 1: “most often” means the tallest stack of marks. Comparing the stacks: three marks above 2, five above $2\frac{1}{2}$, four above 3, two above $3\frac{1}{2}$ and one above 4.

Step 2: the tallest stack is the one above $2\frac{1}{2}$, with five marks.

Step 3: now answer the question that was asked — which *height*, not how many plants. The height that happened most often is $2\frac{1}{2}$ inches.

Watch for: the answer 5. That is how many marks are in the tallest stack, which answers “how many plants”, a different question.

6. Find and fix: Kaya's 5 plants.

Step 1: name what Kaya did. She counted the marked heights along the line — 2, $2\frac{1}{2}$, 3, $3\frac{1}{2}$, 4 — and there are five of them. That counts labels, not plants.

Step 2: each $\times$ is one plant, so add the stacks: $3 + 5 + 4 + 2 + 1$.

Step 3: altogether 15 plants were measured.

Full-credit answer says: “the marks are the plants; the numbers under the line are the heights.” A useful check: fifteen plants in a class is believable, five different heights is not the same fact at all.

7. Find and fix: Sam’s 4th ordered pair.

Step 1 (part a): count along Pattern A: 1, 3, 5, 7. Its 4th term is 7, and A is named first, so 7 belongs in the first slot.

Step 2 (part b): count along Pattern B: 3, 9, 15, 21. Its 4th term is 21, which belongs in the second slot.

Step 3 (part c): the correct 4th ordered pair is (7, 21). Sam wrote (21, 7) — both numbers are right, but he put Pattern B’s term first, so the pair describes a point 21 across and 7 up instead of 7 across and 21 up.

Watch for: a pair written in the wrong order even when both terms are correct. Whichever pattern is named first in (A , B) supplies the first number, every time.

8. Show what you know: (6, 2) and (2, 6).

Open task — flagged for manual review. A model answer is below; accept any equivalent wording.

(a) (6, 2) is plotted 6 steps across from the corner and 2 steps up. (2, 6) is plotted 2 steps across and 6 steps up. The two dots sit on opposite sides of the diagonal, well apart.

(b) Model sentence: “The two numbers do different jobs. The first one counts across and the second counts up, so swapping them sends the point to a different place — the same two numbers, used in the other order.”

(c) Model memory aids: “along the hall, then up the stairs”; “ x comes before y in the alphabet, and across comes before up”; “crawl before you climb”.

What a full-credit answer must contain: both points plotted in the right squares and labelled; a sentence that says what *each* number counts, rather than only “they are different”; and a memory aid that names the order, not just the letters.